

Exercise: Visualizing Amounts

Overview

This exercise provides some practice creating plots to visualize amounts, namely counts and is designed to be completed in 60 minutes or fewer. In particular, you will be tasked with evaluating visualizations to replicate them on your own based on your knowledge of how plots are created in `{ggplot2}`. Although you should have time to complete the parts, if you are unable, *as with all exercises, you are encouraged to complete it outside of class time* so that you are able to incorporate your experiences and knowledge into future exercises. You may need to consult course reading materials located at the course site as some elements may not have been covered in the basic content contained in associated videos.

This exercise assumes you have an understanding of `{dplyr}` functions like `select()`, `filter()`, `mutate()`, and `summarize()` and `{ggplot2}` functions like `ggplot()`, `geom_point()`, `geom_bar()`, and `aes()`.

This exercise focuses on:

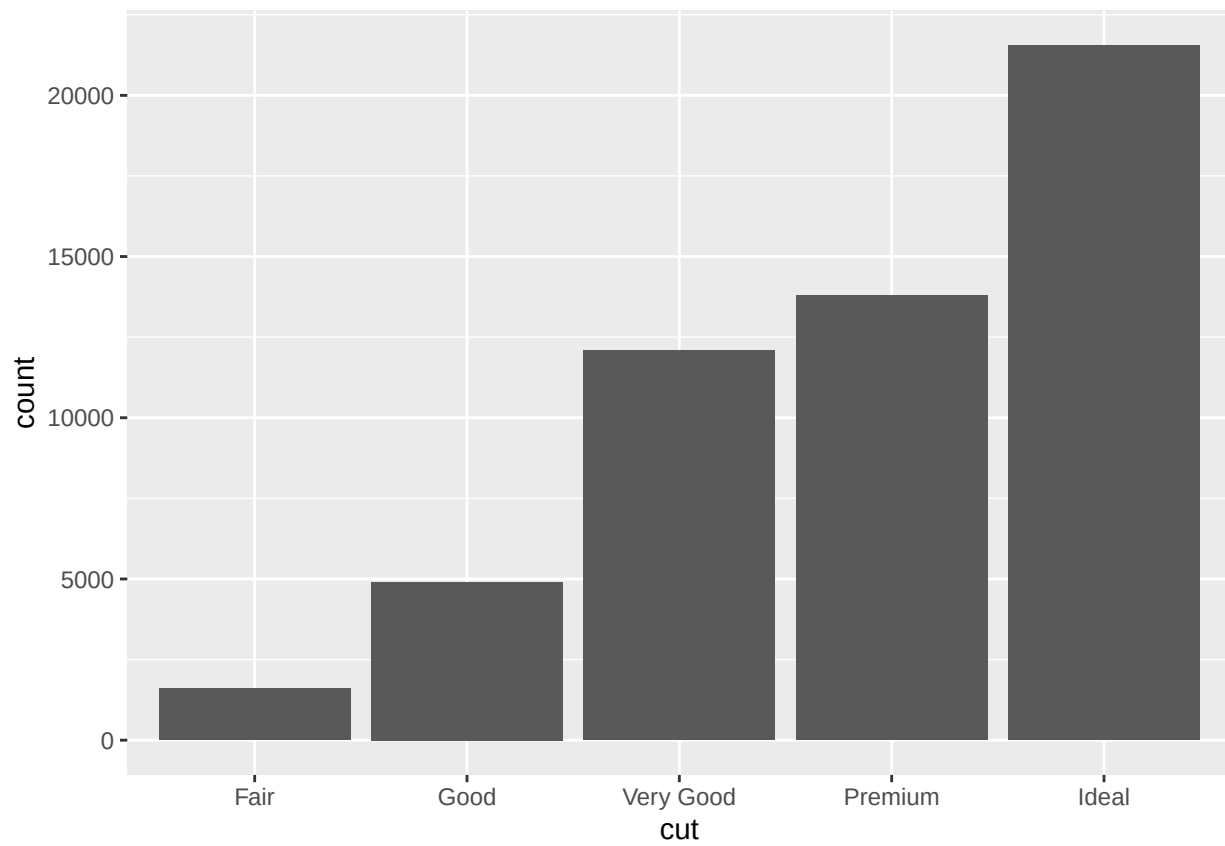
- Creating visualization for amounts
- Adjusting spatial position of geoms
- Mapping variables to aesthetics
- Inspecting data visualizations
- Writing code to replicate plots

You will use `ggplot2::diamonds` for this exercise.

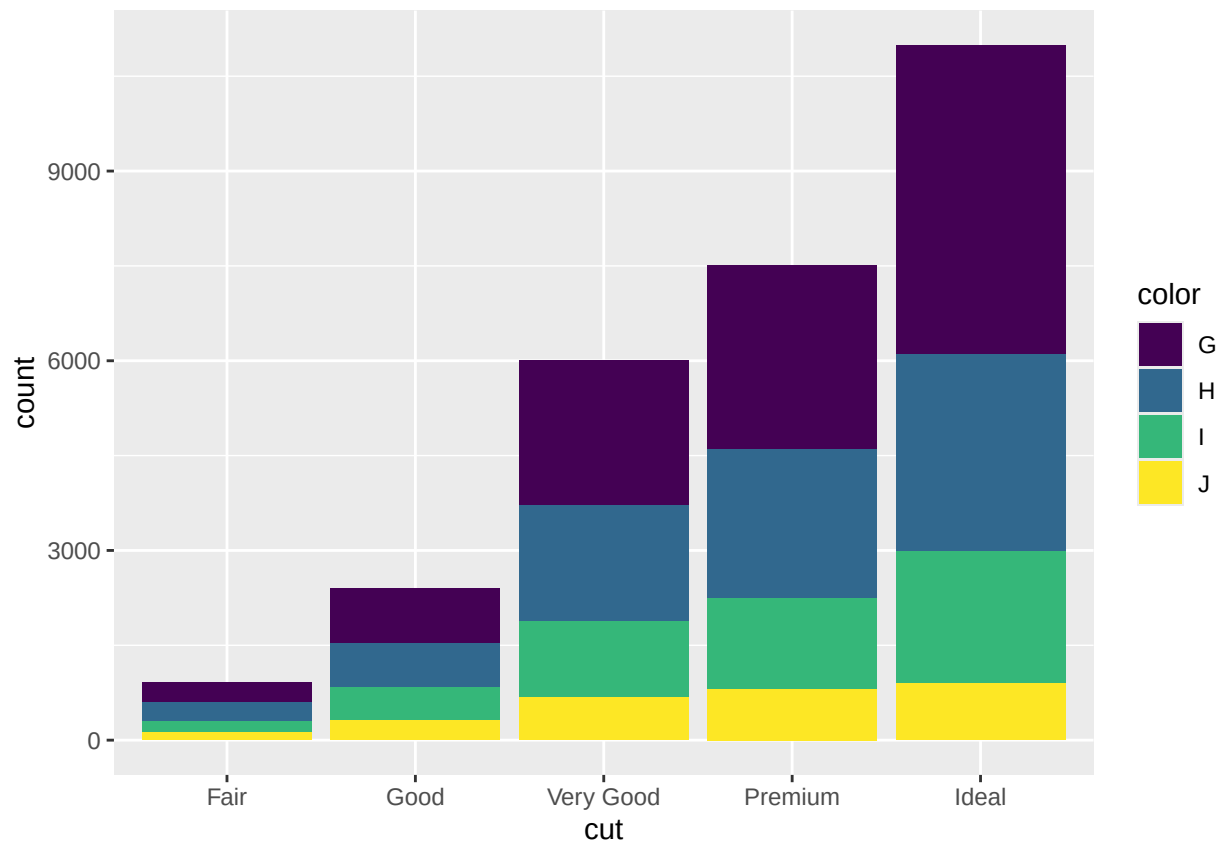
This exercise uses:

- Your RStudio Git version-control project **dataviz-exercises**
- `{here}`, `{dplyr}`, `{ggplot2}` functions and functions from relevant Base R libraries (e.g., `read.csv()`, `readRDS()`, `saveRDS()`, etc.)
- Your knowledge from past in-class exercises, videos, homework, etc. and corresponding modules from the course site.

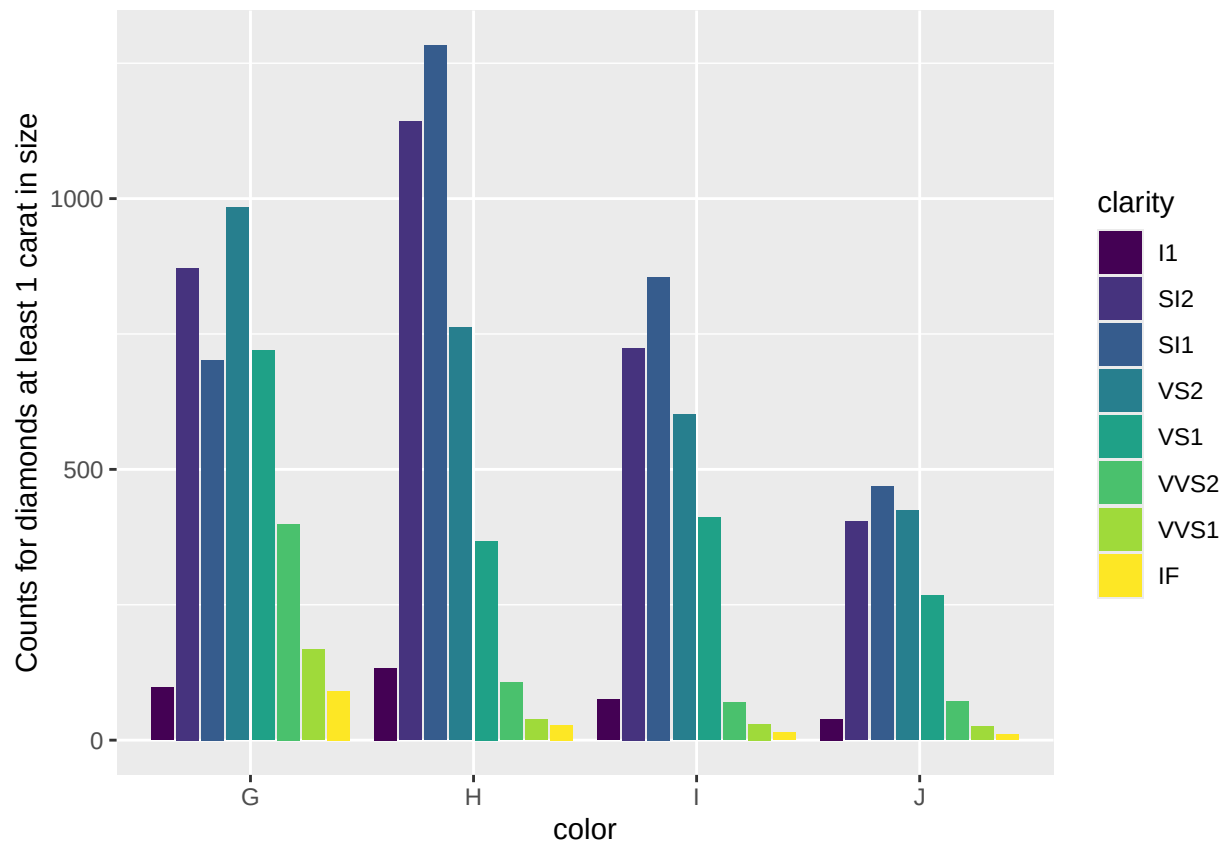
Part 1: Using the relevant libraries, write the code to replicate this plot.



Part 2: Using the relevant libraries, write the code to replicate this plot.



Part 3: Using the relevant libraries, write the code to replicate this plot.



Part 4: Looking Forward: Consider Limitations

Consider limitations with the plots so that you have some goals for interacting with topics on the horizon. Looking at the plots, are there plot elements that could be improved?

- Would other geoms communicate data patterns better?
- Are axes, ticks, and labels clear or could they benefit from something?
- Are colors clear, relevant, biasing?
- Is font size clear, large, enough, or something else?
- Other items?

With any remaining time, discuss with your team tasks that you need to accomplish.